



Recording 2026-10-08 080112.mp4
File

```
import random
```

```
import pygame
```

```
CELL, COLS, ROWS = 20, 30, 30
```

```
WIDTH, HEIGHT = COLS * CELL, ROWS * CELL + 30
```

```
ZONE_TOP = ROWS - 6
```

```
TICK, PLAYER_SPEED, BULLET_SPEED = 0.09, 260, 620
```

```
MUSHROOM_HP = 4
```

```
def mushroom_color(hp):
```

```
    """Return an (r, g, b) colour for a mushroom with the given hit points, or None for the default."""
```

```
    pass
```

```
def on_segment_hit(segment, score):
```

```
    """Called whenever a centipede segment is shot; add sparkles, sounds, or bonus points here."""
```

```
    pass
```

```
...
```

Show more ▾

Yes. Based on the README, there are **4 things to fix/implement**:

1. Fix mushroom durability so it takes **4 hits**, not 3.

2. Implement `mushroom_color`(hp) .
3. Implement `on_segment_hit`(segment, score) .
4. Implement `wave_speed_bonus`(wave) .

I would also make the segment-hit feature visibly work by adding a small **spark effect** when a centipede segment is destroyed.

Here is the complete corrected `game.py` you can copy-paste.

</> Python

```
import random
import pygame

CELL, COLS, ROWS = 20, 30, 30
WIDTH, HEIGHT = COLS * CELL, ROWS * CELL + 30
ZONE_TOP = ROWS - 6
TICK, PLAYER_SPEED, BULLET_SPEED = 0.09, 260, 620
MUSHROOM_HP = 4

# Stores temporary spark effects created when a centipede segment is hit.
HIT_SPARKS = []

def mushroom_color(hp):
    """
    Return a colour based on the mushroom's remaining hit points.

    A mushroom with full HP uses the normal colour.
    As HP decreases, the mushroom becomes more red/yellow,
    making its damage visually obvious.
    """

    if hp == MUSHROOM_HP:
```

```
        return (200, 80, 170)

    if hp == 3:
        return (220, 120, 150)
```

What was actually changed

Task 1 — Mushroom durability

The important bug was:

</> Python

```
if self.mushrooms[cell] <= 1:
```

That removes the mushroom when it reaches 1 , meaning it survives only 3 hits.

It is now:

</> Python

```
if self.mushrooms[cell] <= 0:
```

So:

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```
Initial HP = 4
Hit 1 → 3
Hit 2 → 2
```

```
Hit 3 → 1
```

```
Hit 4 → 0 → mushroom disappears
```

Task 2 — Mushroom colors

`mushroom_color()` now changes the color according to HP:

```
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```

```
4 HP → purple
```

```
3 HP → pink
```

```
2 HP → orange
```

```
1 HP → yellow
```

So you can visibly demonstrate that the mushroom is being damaged without disappearing prematurely.

Task 3 — Segment hit effect

When a segment is shot:

```
</> Python
```

```
on_segment_hit(segment, self.score)
```

creates a temporary spark at the segment's last position.

The spark is then drawn for about 0.25 seconds. This gives you an obvious **before/after feature** to show in your second video.

The existing scoring remains:

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```
Head hit  → +100
Body hit  → +10
```

and the body hit splits the chain into two independent chains.

Task 4 — Wave speed

The implementation from the README is:

</> Python

```
return 1 + 0.15 * (wave - 1)
```

Therefore:

Wave	Speed multiplier
1	1.00×
2	1.15×
3	1.30×
4	1.45×
5	1.60×

Because the game calculates:

</> Python

```
effective_tick = TICK / wave_speed_bonus(self.wave)
```

a higher wave produces a smaller tick interval, so the centipede moves faster.

One important thing for your submission

For the **before video**, you should run the original code first and demonstrate the mushroom bug: shoot the same mushroom repeatedly and show that it disappears on the **third** hit.

For the **after video**, demonstrate:

1. Mushroom survives hits 1–3.
2. Mushroom disappears on hit 4.
3. Damaged mushrooms change color.
4. Shooting a centipede segment produces the spark.
5. Body shot splits the centipede.
6. Clearing a wave starts the next wave.
7. Later waves move faster.

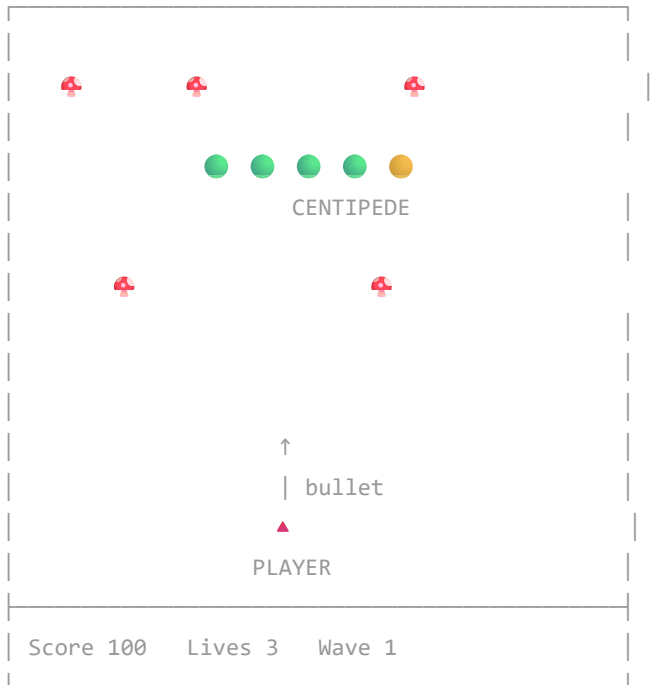
That will directly correspond to the four tasks in the README.

i dont get it what is this game supposed to be doing

Basically, this is a **simple version of the classic arcade game Centipede**.

Think of the screen like this:

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Your job as the player

You control the **blue triangle at the bottom**.

You can:

- Move **left/right**
- Move **up/down**, but only in the bottom section
- Press **Space** to shoot

Your goal is:

Shoot and destroy the entire centipede without losing all your lives.

What is the centipede doing?

The green/yellow circles are the centipede.

It starts at the top:

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and moves horizontally:

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When it reaches the edge:

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it turns around and moves down:

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Then it goes back across the screen.

What do the mushrooms do?

The mushrooms are **obstacles**.

For example:

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When the centipede encounters the mushroom, it changes direction and moves down.

You can also **shoot the mushrooms**.

Each mushroom has **4 HP**:

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 HP = 4

↓ shoot

 HP = 3

↓ shoot

 HP = 2

↓ shoot

 HP = 1

↓ shoot

gone

That's what **Task 1** is testing.

What happens when you shoot the centipede?

Suppose you have:

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and shoot the middle:

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The centipede gets split into two chains:

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These are now **two independent centipedes/chains** that move separately.

That's the main reason the code has things like:

</> Python

```
self.chains
```

and:

</> Python

```
split_chain()
```

Why is the yellow segment special?

The **first** segment of every chain is the head.

So:

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Actually, visually the code makes the head **yellow** and the body **green**.

If you shoot the head, you get:

+100 points

If you shoot a body segment:

+10 points

and the chain splits.

What are the four tasks actually adding?

The original game is already mostly functional.

The lab is basically asking you to **repair and enhance** it.

Task 1 — Fix mushrooms

Currently:

```
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```

```
4 → 3 → 2 → DEAD
```

That's wrong.

You need:

```
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```

```
4 → 3 → 2 → 1 → DEAD
```

Task 2 — Make mushroom damage visible

Instead of the mushroom looking exactly the same, its color changes as you damage it:

```
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```

```
HP 4 → purple
```

```
HP 3 → pink
```

```
HP 2 → orange
```

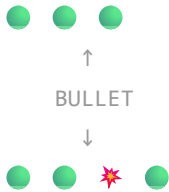
```
HP 1 → yellow
```

```
HP 0 → gone
```

Task 3 — Add an effect when you shoot the centipede

When you hit one:

```
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```



You get a little spark effect.

Task 4 — Make each wave harder

Wave 1:

```
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```

```
Centipede: normal speed
```

Wave 2:

```
</> Plain text
```

```
Centipede: 15% faster
```

Wave 3:

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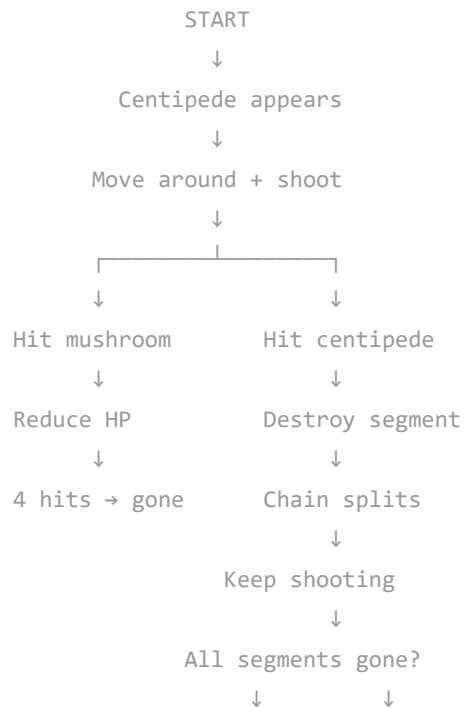
Centipede: 30% faster

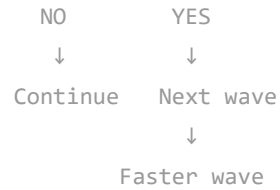
and so on.

The overall game loop

So the game is essentially:

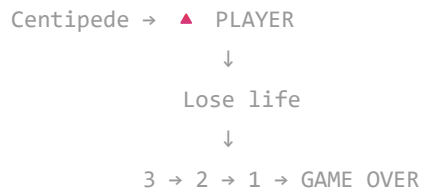
</> Plain text





And if the centipede touches your player:

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You get a short period of **invulnerability after losing a life**, so you aren't immediately hit again.

So if you're wondering "what am I actually supposed to do when I run this?":

Move the blue triangle around at the bottom, press Space to shoot the mushrooms and centipede, destroy the entire centipede, survive, and progress through increasingly faster waves.

i need this entire chat as a pdf how do i do that

If you mean **the entire ChatGPT conversation, including all your messages and my replies**, the easiest way is:

On ChatGPT Web

1. Open this conversation.
2. Scroll through the chat once so everything has loaded.
3. Press **Ctrl + P** on Windows.

4. In the print window, set **Destination** to **Save to PDF**.
5. Click **Save**.
6. Choose where you want to save it.

That should give you a PDF containing the conversation as displayed.

If you want a cleaner PDF

I can also **create a properly formatted PDF of this conversation for you** with:

- Your messages and my responses
- Code formatted in monospace/code blocks
- Headings preserved
- The README/task information included
- A clean layout suitable for submitting or keeping as your lab record

If that's what you want, say "**make the PDF**" and I'll generate it.